MPDD-AVG: Multimodal Personality-Aware Depression Detection via Audio-Visual Interview and Gait Analysis

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***Abstract*—Despite growing interest in automated depression detection, existing methods predominantly rely on conversational modalities while neglecting psychomotor behavioral signatures such as gait, and rarely incorporate individual-level factors including personality traits and demographic characteristics into their modeling frameworks. To bridge these gaps, the MPDD-AVG 2026 challenge introduces a multimodal benchmark integrating audio-visual interview data, wearable gait sequences, and rich personality and demographic annotations across two age cohorts. In this paper, we present the official baseline system employing a multimodal fusion architecture that encodes audio-visual cues, spectral-temporal gait features, and personality-conditioned textual embeddings derived from Big Five traits and demographic attributes. We evaluate the baseline on the MPDD-Young dataset across three tracks: Audio-Visual with Personality (A-V+P), Audio-Visual-Gait with Personality (A-V-G+P), and Gait with Personality (G+P), under both PHQ-9 regression and binary/ternary classification settings. Results demonstrate that gait incorporation yields consistent performance gains, and personality-aware modeling further improves depression assessment, validating the importance of psychomotor and individual-difference modeling for robust, personalized depression detection.**

***Keywords*—**Personality-Aware, Multimodal Depression Detection, Gait Analysis

I. Introduction

DEPRESSION is a leading cause of global disability, imposing substantial societal and economic burdens on millions worldwide [1]. The advent of specialized benchmarks such as AVEC [2] and DAIC-WoZ [3] has significantly advanced audio-visual (A-V) depression detection, yet two critical limitations persist in existing methodologies.

First, current paradigms largely overlook *psychomotor retardation*, a hallmark objective sign of clinical depression [4] that manifests in ambulatory behaviors such as gait kinematics [5]. By confining observations to static interview settings, existing methods fail to capture this clinically established behavioral dimension, leaving a structurally important signal unexploited in computational frameworks. Second, prevailing models assume a uniform behavior-to-symptom mapping, fundamentally neglecting inter-individual heterogeneity driven by personality traits [6], demographic backgrounds, and physical comorbidities [7]. This limitation is particularly pronounced in late-life depression (LLD), where depressive manifestations are closely intertwined with underlying medical burdens and age-specific contextual factors [8].

To bridge these gaps, we introduce the **MPDD-AVG Challenge 2026**, which advances a holistic Audio-Visual-Gait paradigm for personality-aware depression detection. MPDD-AVG systematically integrates semi-structured interview data with spontaneous ambulatory phenotypes captured via wearable inertial measurement unit (IMU) sensors, and is organized into two age-specific cohorts: MPDD-Young and MPDD-Elderly (110 subjects each), enabling explicit modeling of age-sensitive and context-dependent symptom manifestations. Critically, MPDD-AVG provides granular individual-difference annotations to support personalized computational modeling, including PHQ-9/HAMD-24 severity scores, Big Five-10 personality traits, native region demographics for the youth cohort, and socioeconomic and somatic disease annotations for the elderly cohort. This rich annotation scheme empowers participants to investigate the complementary diagnostic value of continuous gait kinematics over static A-V cues, modulated by physiological and contextual priors. The challenge comprises two age-specific tracks, each featuring three sub-tracks: A-V+P, A-V-G+P, and G+P.

In this paper, we present the official baseline system for the MPDD-AVG 2026 challenge, providing a foundational framework and rigorous benchmark for participants. Our contributions are summarized as follows:

- We introduce the MPDD-AVG dataset, the first benchmark to jointly incorporate audio-visual, gait, and personality annotations for depression detection across two demographically distinct age cohorts.
- We propose a multimodal fusion baseline that integrates audio-visual cues, spectral-temporal gait features, and personality-conditioned textual embeddings, establishing competitive reference performance across all challenge tracks.
- We empirically demonstrate that gait modality incorporation yields consistent performance gains, and that personality-aware modeling further improves depression assessment, validating the importance of psychomotor

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TABLE I
COMPARISON OF REPRESENTATIVE DATASETS FOR DEPRESSION-RELATED MULTIMODAL ANALYSIS.

Dataset	A-V	Gait (IMU)	Depression	Personality	Lifespan Cohorts	Comorbidity
DAIC-WoZ	✓	-	✓	-	-	-
Pittsburgh	✓	-	✓	-	-	-
D-Vlog	✓	-	✓	-	-	-
MODMA	✓	-	✓	-	-	-
AMIGOS	✓	-	-	✓	-	-
Twitter-MDD	-	-	✓	✓	-	-
MPDD 2025	✓	-	✓	✓	-	✓
MPDD-AVG (Ours)	✓	✓	✓	✓	✓	✓

and individual-difference modeling for robust depression detection.

II. DATASET

Despite the burgeoning interest in multimodal approaches for depression detection, existing methodologies exhibit notable limitations. While pioneering benchmarks like Pittsburgh [9], D-Vlog [10], CMDC [11], and MODMA [12] have advanced the field, most existing approaches predominantly rely on conversational or interview-based modalities (audio-visual), while ambulatory behavioral signatures such as gait characteristics, though recognized as important clinical indicators of psychomotor symptoms, remain largely underexplored. Moreover, many methods establish direct correlations between behavioral signals and depression levels without explicitly modeling individual differences, failing to account for the nuanced variations influenced by personality traits and demographic factors. While affective computing datasets like AMIGOS [13] or SEED [14] incorporate personality-related signals, they are not designed to jointly model depression, gait, and cohort-specific clinical context in a unified benchmark.

To bridge these gaps, this paper presents a comprehensive baseline model for the MPDD-AVG Challenge, which introduces a novel benchmark integrating semi-structured interviews with continuous gait monitoring. Our work directly addresses the aforementioned limitations. First, we propose a unified multimodal baseline framework that uniquely combines audio, visual, and gait data, enabling a holistic analysis spanning cognitive-linguistic, affective-paralinguistic, and psychomotor domains. Second, to effectively model individual heterogeneity, our framework incorporates personality-derived textual embeddings and demographic information.

Compared to prior works, our baseline is tailored to the unique characteristics of the MPDD-AVG dataset, effectively handling the heterogeneity of multi-source data (interview-based A/V and sensor-based gait) while leveraging granular individual difference annotations. By doing so, this study establishes a solid baseline for personalized depression detection that accounts for age-specific manifestations and inter-individual variability, facilitating the development of more robust and generalizable models in this evolving research direction.

A. MPDD-Young

The MPDD-Young dataset comprises data from 110 college student participants, serving as a core component of the MPDD-AVG 2026 challenge. Extending the previous MPDD dataset [15], this updated version integrates audiovisual interviews with ambulatory gait monitoring to provide a more comprehensive multimodal perspective.

Data Collection and Protocol. Participants first undergo a semi-structured interview assessing academic stress, social functioning, and emotional well-being, during which facial expressions and speech are recorded via a front-facing camera and microphone. Subsequently, participants perform a natural walking task within a designated area while equipped with wearable IMU sensors, capturing continuous gait sequences including temporal and spatial parameters (e.g., stride length, cadence) that serve as clinically relevant indicators of psychomotor symptoms.

Annotations and Supported Tasks. The dataset provides PHQ-9 severity scores, Big Five-10 personality trait evaluations, and demographic information including gender, age, and birth region. Based on these annotations, MPDD-Young supports three tasks: (1) *binary classification* distinguishing non-depressed from depressed individuals; (2) *ternary classification* categorizing participants into normal, mild, and severe depression groups; and (3) *regression* for continuous PHQ-9 score prediction. This comprehensive scheme facilitates investigation of how academic stress, social environment, and personality traits collectively contribute to depression manifestation in young adults.

B. MPDD-Elderly

The MPDD-Elderly dataset comprises 110 older adult participants, forming the geriatric cohort of the MPDD-AVG benchmark. Unlike early-onset depression, late-life depression (LLD) manifestations are heavily influenced by somatic comorbidities and socio-environmental stressors, motivating an annotation scheme that explicitly preserves granular inter-individual differences beyond monolithic behavior-to-symptom mappings.

Data Collection and Protocol. Data acquisition follows a dual-phase protocol to capture complementary behavioral phenotypes. Participants first engage in semi-structured clinical interviews, yielding episodic audio-visual affective cues. To explicitly quantify psychomotor retardation, participants subsequently perform unconstrained walking tasks equipped with wearable IMU sensors. This dual-session design bridges static conversational behaviors with continuous ambulatory kinematics, enabling holistic characterization of depressive symptomatology.

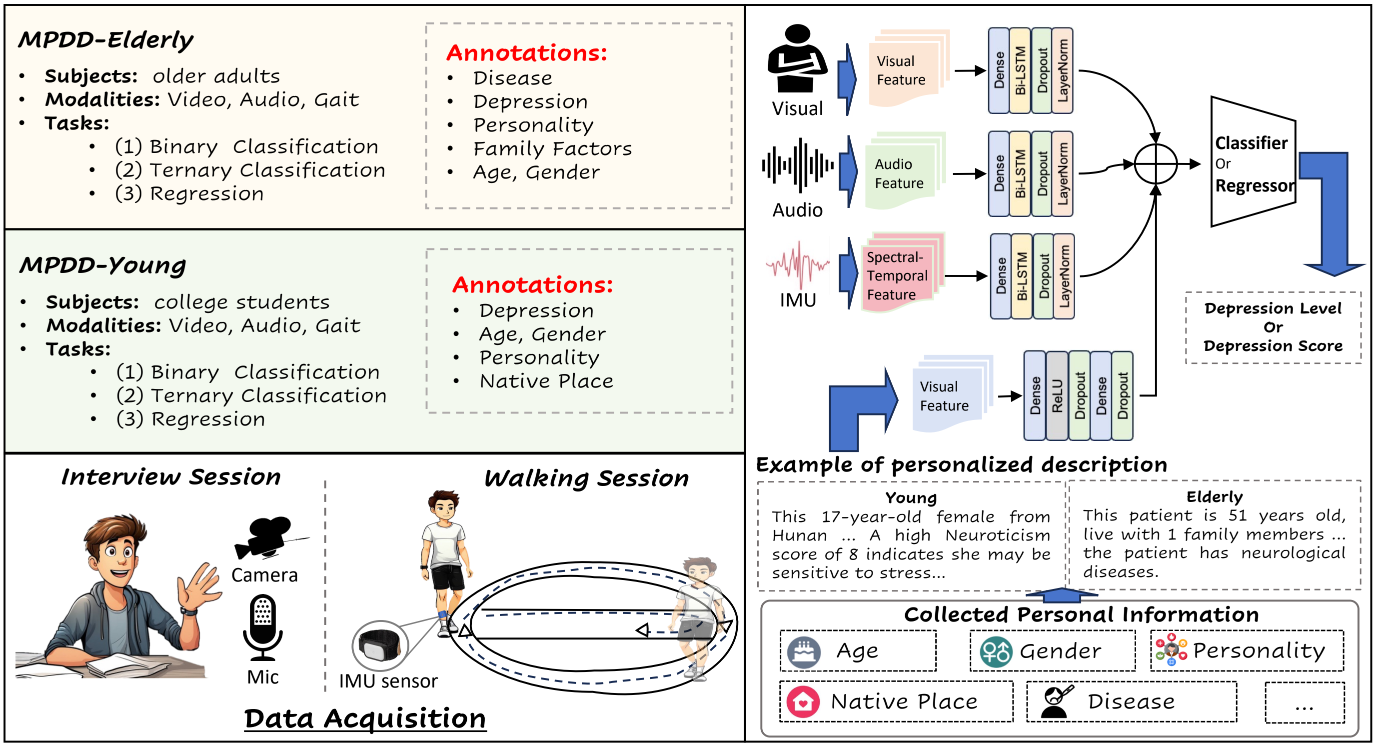

Fig. 1. Overview of the MPDD Dataset and Baseline System. (Left) The MPDD benchmark includes MPDD-Elderly (older adults) and MPDD-Young (college students), which are collected via interview sessions (video + audio) and walking sessions (IMU/gait). Each sample is annotated with depression labels, personality traits, and demographic information, supporting binary classification, ternary classification, and regression tasks. Personalized textual descriptions are generated from the collected metadata to provide subject-specific context. (Right) The baseline system extracts visual, acoustic, and spectral-temporal features from the three modalities, encodes each through a dedicated Dense–BiLSTM–Dropout–LayerNorm module, and fuses the representations to predict depression level or score.

Annotations and Supported Tasks. MPDD-Elderly provides multidimensional annotations to model cohort heterogeneity. Depression severity is annotated via PHQ-9 scores, supporting three tasks: (1) *binary classification* of non-depressed versus depressed individuals; (2) *ternary classification* into normal, mild, and severe groups; and (3) *regression* for continuous severity score prediction. The dataset is further enriched with Big Five-10 personality profiles, family factors including financial stress and co-resident counts, and somatic disease annotations stratifying healthy individuals against those with endocrine, circulatory, and neurological disorders. These structured annotations compel the design of architectures that explicitly condition behavioral representations on a patient's physiological and socioeconomic profile, advancing LLD assessment toward highly individualized symptom decoding.

TABLE II
THE DISTRIBUTION OF DIFFERENT PHQ-9 SCORE LABELS IN TRACK 1 AND TRACK 2.

Dataset	Label	Training set	Testing set	Total
Track 1 (Elderly)	0 (Normal)	44	11	55
	1 (Mild)	33	8	41
	2 (Severe)	11	3	14
Track 2 (Young)	0 (Normal)	41	10	51
	1 (Mild)	37	10	47
	2 (Severe)	10	2	12
Total		176	44	220

III. Baseline System

A. Model

To establish a rigorous and reproducible benchmark for MPDD-AVG, we introduce a canonical multimodal baseline architecture inspired by general multimodal learning principles [16] and shared across all sub-tracks (`A-V+P`, `A-V-G+P`, and `G+P`). Rather than engineering heavily specialized networks for individual modality settings, this unified framework is designed to perform depression classification under a shared modeling protocol. This design effectively isolates the diagnostic contributions of different behavioral combinations and individual-difference priors, ensuring that comparisons across sub-tracks—and between the young and elderly cohorts—remain strictly controlled.

The architecture processes two fundamentally distinct streams of information. The first is modality-specific sequential evidence, encompassing episodic audio-visual dynamics extracted from semi-structured interviews and continuous ambulatory kinematics from IMU-based gait tracking. The second is participant-level contextual information, encoded as a fixed-length embedding derived from the released individual-difference attributes. For any given sub-track, the model activates only the specified input modalities while preserving a consistent processing pipeline.

To preserve modality-specific temporal dynamics before multimodal integration, each behavioral stream is encoded independently. Audio, video, and gait sequences are mapped into compact latent representations via lightweight temporal encoders. Concurrently, the participant-level embedding is transformed by a shallow projection module into a compatible contextual prior. By maintaining this consistent temporal encoding and late-fusion logic, the baseline successfully retains sequence-level dynamics in a streamlined framework, circumventing the architectural bloat typically associated with multi-stream networks. More broadly, this design is also compatible with attention-based fusion paradigms derived from Transformer architectures [17].

Upon modality-specific encoding, the resulting representations are aggregated at a high level and fed into a shallow classification head to predict depression severity. Under `A-V+P`, the model fuses conversational cues with participant context. Under `A-V-G+P`, ambulatory psychomotor evidence (Gait) is further integrated. Under `G+P`, the framework isolates the joint diagnostic yield of gait kinematics and individual differences. This unified prediction logic establishes a transparent reference for evaluating multimodal fusion strategies.

B. Feature Representations

Deviating from conventional fixed-window aggregation strategies, the released baseline adopts an **unsegmented sequence-level** feature extraction paradigm across all modalities. By retaining frame-level or sequence-level descriptors without prior windowing, the pipeline compels downstream encoders to dynamically learn global temporal dependencies. Consequently, audio and video preserve episodic conversational shifts, gait retains continuous kinematic rhythms, and individual differences are instituted as a stable subject-level context. This representation strategy aligns perfectly with the shared sequential modeling framework.

- **Acoustic Representations**: We provide three complementary feature families. MFCCs capture low-level spectral and timbral structures, offering a classical characterization of speech acoustics. OpenSMILE [18] extracts hand-crafted paralinguistic descriptors, quantifying nuanced prosodic and voice-quality patterns fundamental to affective computing. Wav2Vec2 [19] supplies deep, pretrained contextualized speech representations, complementing traditional descriptors with high-level semantic abstractions.
- **Visual Representations**: The baseline incorporates DenseNet [20], ResNet [21], and OpenFace extractors. DenseNet and ResNet provide robust frame-level appearance embeddings of the upper body and face within a generic visual space. Conversely, OpenFace explicitly targets fine-grained facial behaviors and expression-related dynamics. This configuration enables a comprehensive contrast between general-purpose spatial features and behavior-oriented facial kinematics.
- **Gait (IMU) Representations**: Serving as an objective proxy for psychomotor retardation, gait is represented by IMU-based motion sequences derived from 3D acceleration, angular velocity, and angle-related orientation signals. These sequences continuously quantify ambulatory dynamics, uniquely capturing sub-clinical movement manifestations that are structurally invisible in stationary conversational expressions.
- **Personalized Description Embeddings**: To computationally operationalize individual-difference metadata, we employ a semantic textualization strategy. Cohort-specific structured attributes (e.g., personality traits, comorbidities) are organized with a template and feed into ChatGLM to yield coherent textual descriptions, which are subsequently encoded into dense, fixed-length semantic vectors. This transforms isolated metadata fields into a stable subject-level contextual prior, explicitly compelling the model to account for inter-individual heterogeneity.

C. Personalized Features

To model individual heterogeneity, we adopt the personalized feature extraction pipeline established in the previous MPDD Challenge [15]. This approach leverages participant-specific attributes, including Big Five-10 personality traits, financial stress levels, and gait-derived psychomotor signatures, to generate individualized representations.

Specifically, we transform these structured attributes into natural language descriptions via a generative language model (ChatGLM3) [22]. These descriptions are then encoded using the RoBERTa-large model [23], from which we extract the hidden state of the first token to obtain a 1024-dimensional embedding. These embeddings serve as compact, semantically-rich personalized features, which are subsequently concatenated with acoustic, visual, and gait features for downstream classification.

IV. METRICS

To comprehensively evaluate the performance of our proposed baseline model, we employ a set of standard metrics for both classification and regression tasks. Specifically, we utilize Accuracy and F1-score for classification, and Concordance Correlation Coefficient (CCC) for regression. Given the potential class imbalance in the dataset, we report both weighted and unweighted (macro) averages for the classification metrics to ensure a robust assessment of the model's capabilities across different depression severity levels.

Accuracy measures the proportion of correct predictions:

$$\text{Acc} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Samples}} \tag{1}$$

The F1-score, as the harmonic mean of precision and recall, is particularly vital for addressing class imbalance frequently encountered in mental health datasets:

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{2}$$

where

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

Macro-F1 computes the F1-score for each class independently and takes the arithmetic mean, ensuring that all depression severity levels contribute equally to the final metric irrespective of their sample frequency:

$$\text{Macro-F}_1 = \frac{1}{C}\sum_{i=1}^{C} F_1^{(i)} \tag{4}$$

where C is the number of classes.

Cohen's Kappa quantifies inter-rater agreement by correcting for chance, providing a robust measure for evaluating clinical diagnostic consistency:

$$\kappa = \frac{p_o - p_e}{1 - p_e} \tag{5}$$

where p_o is the observed agreement and p_e is the expected agreement by chance.

For the depression severity prediction task, we employ: Root Mean Squared Error (RMSE) quantifies the magnitude of prediction errors:

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \tag{6}$$

Mean Absolute Error (MAE) provides an interpretable average prediction error:

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i| \tag{7}$$

The Concordance Correlation Coefficient (CCC) evaluates the agreement between predicted and actual PHQ-9 scores by combining measures of precision and accuracy [24]:

$$\text{CCC} = \frac{2\rho\,\sigma_y\,\sigma_{\hat{y}}}{\sigma_y^2 + \sigma_{\hat{y}}^2 + (\mu_y - \mu_{\hat{y}})^2} \tag{8}$$

where ρ is Pearson's correlation coefficient, σ denotes the standard deviation, and μ denotes the mean.

The final evaluation score for each track is computed as the weighted average across all classification and regression tasks:

$$\text{Score}_{\text{track}} = \alpha \cdot \text{Macro-F}_1 + \beta \cdot \text{CCC} + \gamma \cdot \kappa \tag{9}$$

where α, β, and γ are weights that sum to 1, reflecting the relative importance of different aspects of model performance. This composite metric ensures comprehensive assessment spanning depression severity classification, continuous score prediction, and diagnostic consistency.

V. Results

The baseline results for the Elderly and Young tracks are presented in Table III. We analyze performance across sub-tracks and tasks to assess the contribution of each modality and the effect of personality-aware modeling.

Effect of Gait Modality. Comparing A-V+P and A-V-G+P sub-tracks reveals consistent performance trends upon incorporating gait features. For the Elderly track, the A-V-G+P sub-track (using mfcc64 + resnet) achieves a binary F1 of **0.6349** and an ACC of **0.6522**, with a Kappa of 0.2756. While the binary performance slightly varies depending on the specific feature backbone used (e.g., wav2vec vs. mfcc64), the integration of multiple modalities remains competitive. For the Young track, incorporating gait yields a significant improvement, with the binary F1 rising to **0.7273** and ACC to **0.7273** in the A-V-G+P sub-track, compared to **0.6812** F1 in the A-V+P sub-track. These results confirm that gait kinematics encode clinically meaningful psychomotor signals that complement audio-visual cues for depression detection.

Gait-Only Modeling. The G+P sub-track, which relies solely on gait and personality inputs, achieves surprisingly strong performance. For the Young track, G+P attains a binary F1 of **0.7684** and ACC of **0.7727**, actually surpassing the multimodal A-V-G+P track. This suggests that ambulatory behavioral patterns alone carry substantial discriminative capacity for depression detection in younger populations. For the Elderly track, G+P yields a binary F1 of **0.5068**, which is lower than the audio-visual fusion tracks (**0.6538** F1), reflecting the greater complexity of psychomotor manifestations in older adults.

Binary vs. Ternary Classification. Across all sub-tracks and both cohorts, ternary classification consistently yields lower F1 and higher error metrics (RMSE/MAE) compared to binary classification. For instance, in the Young A-V-G+P sub-track, binary F1 reaches **0.7273** while ternary F1 drops to **0.4900**, underscoring the challenge of fine-grained severity stratification (distinguishing mild from severe states) compared to simple detection.

Cohort-Level Comparison. The Young track generally achieves higher performance across all metrics than the Elderly track. For example, the highest binary F1 in the Young track reaches **0.7684** (G+P), whereas the Elderly track peaks at **0.6538** (A-V+P). This likely results from the more homogeneous demographic profile of the younger cohort, whereas the Elderly group exhibits greater variability due to the confounding influence of somatic health and diverse socioeconomic backgrounds.

VI. Conclusion

In this work, we present the official baseline system for the MPDD-AVG 2026 Challenge, introducing a comprehensive benchmark for personality-aware, multimodal depression detection across two age cohorts. By jointly integrating audio-visual interview cues, continuous gait kinematics captured via wearable IMU sensors, and granular individual-difference annotations including personality traits and demographic attributes, the MPDD-AVG dataset directly addresses two critical limitations of existing benchmarks: the neglect of psychomotor behavioral signatures and the absence of explicit individual heterogeneity modeling. Our baseline establishes transparent and reproducible reference performance across all three sub-tracks (A-V+P, A-V-G+P, and G+P) under both PHQ-9 regression and binary/ternary classification settings. Empirical results consistently demonstrate that gait incorporation yields substantial diagnostic gains over audio-visual-only baselines, while personality-conditioned modeling further improves depression assessment across both cohorts. These findings underscore the clinical and computational value

TABLE III
BASELINE RESULTS ON MPDD DATASET (TRACK 1: ELDERLY, TRACK 2: YOUNG). THE BEST RESULTS ARE REPORTED.

Track	Sub-track	A/V Features	Task	F1	ACC	Kappa	CCC	RMSE	MAE
Elderly	A-V+P	mfcc64 + densenet	Binary	0.6057	0.6087	0.2129	0.0316	4.0273	3.1298
Elderly	A-V+P	mfcc64 + resnet	Ternary	0.3167	0.4348	0.1128	0.0748	3.9666	2.7537
Elderly	A-V-G+P	mfcc64 + resnet	Binary	0.6349	0.6522	0.2756	0.0541	4.1305	3.0189
Elderly	A-V-G+P	opensmile + resnet	Ternary	0.3921	0.5217	0.2215	0.0758	3.9414	2.5122
Elderly	G+P	-	Binary	0.5068	0.5217	0.0996	-0.0007	4.0217	2.9532
Elderly	G+P	-	Ternary	0.2020	0.4348	0.0000	0.0023	4.8645	2.8421
Young	A-V+P	mfcc64 + resnet	Binary	0.6812	0.6818	0.3636	0.0842	4.4054	3.0120
Young	A-V+P	wav2vec + resnet	Ternary	0.4577	0.6364	0.3600	0.0231	4.5909	3.0993
Young	A-V-G+P	wav2vec + resnet	Binary	0.7273	0.7273	0.4546	0.0322	4.5861	3.1134
Young	A-V-G+P	mfcc64 + resnet	Ternary	0.4900	0.6818	0.4211	0.0407	4.7370	3.0461
Young	G+P	-	Binary	0.7684	0.7727	0.5456	0.0829	4.4526	2.9220
Young	G+P	-	Ternary	0.4731	0.5909	0.3125	0.1536	4.5388	3.0635

of extending depression detection beyond static conversational observations toward a holistic, personalized multimodal paradigm. We hope MPDD-AVG serves as a foundational testbed to advance automated depression assessment research, encouraging the community to develop models that are more sensitive to psychomotor symptoms and more robust to inter-individual variability in depressive symptomatology.

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